

AC2205 Marking Scheme

Intermediate Management Accounting · Summer 2025 · Paper 1

Examination Session and Year	Summer 2025
Module Code	AC2205
Module Title	Intermediate Management Accounting
Paper Number	1
External Examiner	Professor Martin Quinn
Head of School/Department	Professor Michelle Carr
Internal Examiners	Dr. Conor Loughnane
Duration of Paper	1.5 hours
Special Requirements	Students are allowed to use calculators that have standard arithmetic, scientific, graphical or algebraic functions provided that they are NOT programmable and can NOT store text (other than algebraic notation).

Instructions to Candidates (as per exam paper): Answer two questions only. All questions carry equal marks (50 each). Show all your workings.

Own-figure rule: where a candidate makes an error at an early stage of a multi-step numerical question, full method marks are awarded for correctly applying the required technique to their own (incorrect) figure in all subsequent steps. Marks are never cascaded away for a single early error.

Marking philosophy: this is a closed-book, time-pressured examination. Academic citations are never required. Discursive answers are marked on depth of explanation and application to the specific scenario, not generic textbook recall.

Question Overview

Question	Scenario	Topics	Marks
Question 1	Bloom & Co	Cash budgeting (receipts and payments), budget commentary, incremental vs rolling budgets	50
Question 2	PureGlow Skincare Ltd	Limiting factor (key factor) analysis, production planning under a resource constraint, relevant costing, joint products/sunk/opportunity cost	50
Question 3	EcoGlow Ltd	Flexible budgeting, flexible budget and volume variances, performance evaluation, criticisms of budgeting	50

Candidates answer any TWO of the three questions above; each is worth 50 marks. This marking scheme covers all three questions in full.

QUESTION 1**Bloom & Co**

Total: 50 marks

Bloom & Co supplies eco-friendly gardening products to retailers across Ireland. The CEO, Alan Green, wants to ensure sufficient cash flow over the coming months. Candidates are required to (a) prepare a cash budget for Quarter Four (October, November, December), (b) comment on the resulting cash position with appropriate advice, and (c) differentiate between incremental and rolling budgets.

Part	Requirement	Marks
(a)	Cash budget for Q4 (Oct, Nov, Dec), showing monthly and quarterly cash position	30
(b)	Commentary on the cash budget, with advice to management	10
(c)	Differentiate between incremental budgets and rolling budgets	10

Question 1(a)**30 marks**

Bloom & Co, budgeted figures July–December (€): Purchases, Sales, Salaries, Overheads, and Advertising given monthly. Cash sales are 40% of total sales (5% discount applied); credit sales (60%) settled 50% in month of sale, 30% one month after, 20% two months after. Credit purchases are 50% of total purchases, paid 70% one month after, 20% two months after, 10% three months after (cash purchases, the other 50%, are paid in the month of purchase). Salaries paid one month in arrears; advertising paid two months in arrears. Overheads (including a weekly depreciation charge of €290, four weeks per month) paid one month in arrears. A new vehicle costing €58,000 is bought in October, paid in four equal monthly instalments starting the month after purchase; an old vehicle is sold in December for €600. Tax payments of €3,900 (September) and €15,200 (October) are expected. A €1,400 sustainability grant is expected in November. The bank balance at the end of September is estimated at €7,000. (a) Prepare a cash budget for Quarter Four, showing the firm's cash position at the end of each month AND for the quarter as a whole.

Full Model Answer - Preliminary workings**Working 1 - cash receipts from sales**

Item	October (€)	November (€)	December (€)
Cash sales this month, net of 5% discount (40% of month's sales × 95%)	74,100	63,840	58,330
Credit sales settled in month of sale (60% of month's sales × 50%)	58,500	50,400	46,050
Credit sales settled one month after sale (prior month's credit sales × 30%)	62,640	35,100	30,240
Credit sales settled two months after sale (two months prior credit sales × 20%)	32,160	41,760	23,400
Total cash receipts from sales	227,400	191,100	158,020

Working detail (October shown; November/December follow the same pattern): October cash sales: $40\% \times €195,000 = €78,000$, less 5% discount = €74,100. October credit sales settled this month: $60\% \times €195,000 \times 50\% = €58,500$. Settled one month after (from September's credit sales): $60\% \times €348,000 \times 30\% = €62,640$. Settled two months after (from August's credit sales): $60\% \times €268,000 \times 20\% = €32,160$.

Full Model Answer (continued) – Cash payments**Working 2 – cash payments for purchases**

Item	October (€)	November (€)	December (€)
Cash purchases this month (50% of month's purchases, paid same month)	44,000	41,250	38,000
Credit purchases paid one month after (prior month's credit purchases × 70%)	49,700	30,800	28,875
Credit purchases paid two months after (× 20%)	12,600	14,200	8,800
Credit purchases paid three months after (× 10%)	4,900	6,300	7,100
Total cash payments for purchases	111,200	92,550	82,775

Assumption stated: Cash purchases (50% of the total) are assumed paid in the month of purchase, consistent with the treatment of cash sales; only credit purchases (the other 50%) follow the stated 70%/20%/10% deferred payment pattern.

Working 3 – salaries, advertising and overheads (payment timing)

Item	October (€)	November (€)	December (€)
Salaries (one month in arrears: Sep, Oct, Nov salary paid)	39,000	41,500	38,500
Advertising (two months in arrears: Aug, Sep, Oct advertising paid)	13,000	14,800	9,500
Overheads, excl. depreciation, one month in arrears (see Working 4)	37,640	43,740	42,340

Working 4 – overheads excluding non-cash depreciation

Item	Value
Weekly depreciation charge	€290
Monthly depreciation (4 weeks × €290)	€1,160
Cash overheads generated in September (€38,800 – €1,160), paid in October	€37,640
Cash overheads generated in October (€44,900 – €1,160), paid in November	€43,740
Cash overheads generated in November (€43,500 – €1,160), paid in December	€42,340

Why depreciation is excluded: Depreciation is a non-cash accounting charge; it must be stripped out of the overheads figure before determining the actual cash payment made the following month, even though it is included within the €290/week figure embedded in the budgeted overheads total.

Full Model Answer (continued) - Other items and full cash budget**Working 5 - other cash items**

Item	October (€)	November (€)	December (€)
New vehicle instalment (€58,000 ÷ 4, starting the month after October's purchase)	0	14,500	14,500
Sale of old vehicle (cash inflow)	0	0	600
Tax payment	15,200	0	0
Sustainability grant received (cash inflow)	0	1,400	0

Vehicle purchase timing: The €58,000 vehicle is bought in October but, per the question, payments are spread over four equal monthly instalments *beginning the month after purchase* — so the first instalment falls in November, not October. No cash payment for the vehicle therefore appears in October at all.

Full cash budget, Quarter Four (October, November, December)

Item	October (€)	November (€)	December (€)	Quarter total (€)
Receipts				
Cash receipts from sales	227,400	191,100	158,020	576,520
Sustainability grant	0	1,400	0	1,400
Sale of old vehicle	0	0	600	600
Total receipts	227,400	192,500	158,620	578,520
Payments				
Payments for purchases	111,200	92,550	82,775	286,525
Salaries	39,000	41,500	38,500	119,000
Overheads (excl. depreciation)	37,640	43,740	42,340	123,720
Advertising	13,000	14,800	9,500	37,300
Tax payment	15,200	0	0	15,200
Vehicle instalments	0	14,500	14,500	29,000
Total payments	216,040	207,090	187,615	610,745
Net cash flow for the month	11,360	(14,590)	(28,995)	(32,225)
Opening cash balance	7,000	18,360	3,770	7,000
Closing cash balance	18,360	3,770	(25,225)	(25,225)

Reconciliation: The quarter-as-a-whole net cash flow of €(32,225) equals the sum of the three monthly net cash flows (€11,360 – €14,590 – €28,995), and the closing balance for the quarter (€(25,225)) equals the opening balance at 1 October (€7,000) plus this net movement, matching the final closing balance at the end of December.

Mark Allocation

Marks	Criteria
26-30	All receipts (sales, grant, vehicle sale) and all payments (purchases, salaries, overheads net of depreciation, advertising, tax, vehicle instalments) correctly calculated and correctly timed; monthly and quarterly closing balances all correct and reconciling.
19-25	Substantially correct with minor timing or computational errors in one or two items (e.g. depreciation not excluded from overheads, or vehicle instalment timing).
10-18	Core structure of the cash budget understood; several timing errors (e.g. cash purchases not assumed paid in month of purchase, or credit sales/purchases percentages misapplied).
0-9	Limited understanding of cash budgeting; minimal correct workings.

Question 1(b)**10 marks**

Comment upon the cash budget that you have produced in part (a), offering whatever advice to the manager that you consider appropriate.

Full Model Answer**Overall assessment**

The cash budget reveals a deteriorating and, by the end of the quarter, seriously concerning cash position: Bloom & Co moves from a healthy €18,360 cash balance at the end of October to an overdraft of €25,225 by the end of December — a swing of over €43,500 in just two months.

What is driving the deterioration

Sales are falling sharply through Q4 (from €348,000 in September to €153,500 in December), which steadily reduces cash receipts even as many payments (purchases, salaries, overheads) remain comparatively high due to the payment-timing lags built into the business's terms. This is compounded by two lumpy cash outflows landing in the same window: the €15,200 tax payment in October and the new €14,500-per-month vehicle instalments beginning in November, which together add significant pressure precisely as sales revenue is declining.

Advice to Alan Green

1. Arrange an overdraft facility or short-term financing with the bank *in advance* of the projected December shortfall, rather than reacting after the fact — the budget gives clear early warning. **2.** Consider whether the timing of the vehicle purchase/instalment schedule could be renegotiated (e.g. a longer instalment period, or delaying the purchase) to smooth the cash impact. **3.** Investigate the cause of the steep sales decline into Q4 and consider promotional activity or seasonal pricing to soften it. **4.** Review credit control on the 20% of credit sales collected two months after sale — tightening this collection period would materially improve near-term cash receipts. **5.** Continue to monitor and update the cash budget monthly (or consider a rolling budget, see part (c)) so early warning signs are not missed as actual conditions unfold.

Mark Allocation

Marks	Criteria
9-10	Correctly identifies the deteriorating trend and the overdraft position by December; explains the key drivers (declining sales, tax payment, vehicle instalments); gives specific, actionable advice (financing, renegotiation, credit control, monitoring).
6-8	Correctly identifies the overdraft position with reasonable explanation of drivers; advice is more generic.
3-5	Some relevant commentary but lacks specific reference to the figures produced in part (a).
0-2	Minimal or no meaningful commentary.

Question 1(c)**10 marks**

Differentiate between "incremental budgets" and "rolling budgets".

Full Model Answer**Incremental budget**

An incremental budget is prepared by taking the prior period's budget (or actual results) as a starting base and adjusting it incrementally for anticipated changes, such as expected inflation or planned growth in activity, rather than building the budget from zero each period. It is quick and straightforward to prepare, but it risks simply carrying forward any inefficiencies or budgetary slack built into the base period, without requiring managers to justify each line of spend afresh.

Rolling (continuous) budget

A rolling budget is continuously updated by adding a new period (e.g. a further month or quarter) onto the end of the budget as each period elapses and is dropped off, so the organisation always maintains a budget covering a full period ahead (for example, a twelve-month rolling budget). This requires more frequent preparation and administrative effort than a budget fixed for a full year in advance, but keeps the budget more current and responsive to changing conditions — particularly valuable for a business like Bloom & Co facing volatile, seasonally declining sales, where a cash budget prepared once a year would quickly become out of date.

Key distinction: The incremental/rolling distinction is about how the *content* of the budget is derived (from a prior base, adjusted incrementally, vs built continuously) and is a separate concept from a fixed budget vs a flexible budget (the distinction tested in Question 3), which is about how a budget is *compared against actual results* after the fact.

Mark Allocation

Marks	Criteria
8-10	Both terms accurately and fully defined, with a clear explanation of the trade-off (speed/simplicity vs currency/responsiveness) between them, ideally applied to Bloom & Co's context.
5-7	Both terms defined with reasonable accuracy; trade-off or application less developed.
2-4	One term defined well; the other only partially, or the distinction between them unclear.
0-1	No meaningful explanation of either term.

QUESTION 1 TOTAL — 50 MARKS

QUESTION 2

PureGlow Skincare Ltd

Total: 50 marks

PureGlow Skincare Ltd manufactures four organic skincare products and faces an anticipated shortage of essential plant extracts in 2025, limiting total supply to 70,000 grams. Candidates are required to (a) formulate a profit-maximising production plan under this constraint, (b) outline four actions to address the demand shortfall, (c) outline the three criteria for a relevant cost, and (d) define three key terms.

Part	Requirement	Marks
(a)	Profit-maximising production plan under the plant extract constraint, with assumptions and commentary	20
(b)	Four potential actions to meet total demand in 2025	12
(c)	Three criteria for a relevant cost/benefit	9
(d)	Define joint products, sunk cost, and opportunity cost	9

Question 2(a)**20 marks**

PureGlow Skincare Ltd, 2025 budgeted unit information for four products (Radiance, HydraGlow, ClearSkin, YouthRenew): sales price, direct material costs (€15/gram of essential plant extract), direct labour, production overheads (50% variable), marketing costs (50% variable), other costs (40% fixed), and sales demand given for each. Total supply of essential plant extract is limited to 70,000 grams. (a) Formulate a production plan that maximises profits in 2025, considering the constraints provided. Clearly state any assumptions made and briefly comment on the production plan developed.

Full Model Answer**Step 1 - confirm the essential plant extract is a genuine limiting factor**

Grams of extract required per unit = direct material cost per unit ÷ €15 per gram. Total grams required to meet full budgeted demand for all four products is calculated below and compared against the 70,000 gram supply.

Working - grams required per unit and to meet full demand

Product	DM cost/unit (€)	Grams/unit (÷€15)	Sales demand (units)	Total grams needed
Radiance	25.00	1.667	18,000	30,000
HydraGlow	40.00	2.667	12,000	32,000
ClearSkin	50.00	3.333	7,500	25,000
YouthRenew	70.00	4.667	4,000	18,667
Total grams needed for full demand				105,667

Constraint confirmed: 105,667 grams would be needed to meet full demand for all four products, against a supply of only 70,000 grams. The essential plant extract is therefore a genuine limiting (key) factor, and production must be prioritised across products to maximise profit from the available supply — simply producing to meet demand in any order is not optimal.

Full Model Answer (continued) - Step 2: contribution per unit of limiting factor

Direct materials and direct labour are wholly variable. Half of production overheads and half of marketing costs are variable; 60% of other costs are variable (40% fixed). Variable cost per unit therefore = DM + DL + 50% of production overheads + 50% of marketing + 60% of other costs.

Working - variable cost and contribution per unit

Product	Radiance	HydraGlow	ClearSkin	YouthRenew
Sales price	75.00	95.00	120.00	150.00
Direct materials	25.00	40.00	50.00	70.00
Direct labour	10.00	12.00	16.00	20.00
Variable production overheads (50%)	4.00	5.00	7.00	9.00
Variable marketing costs (50%)	2.50	3.50	5.00	6.00
Variable other costs (60%)	7.20	9.00	10.80	13.20
Total variable cost	48.70	69.50	88.80	118.20
Contribution per unit	26.30	25.50	31.20	31.80

Working - contribution per gram of essential plant extract (the limiting factor), and ranking

Product	Contribution/unit (€)	Grams/unit	Contribution/gram (€)	Rank
Radiance	26.30	1.667	15.78	1st
HydraGlow	25.50	2.667	9.56	2nd
ClearSkin	31.20	3.333	9.36	3rd
YouthRenew	31.80	4.667	6.81	4th

Key point on ranking: Ranking products by contribution *per unit* alone would be misleading here — YouthRenew has the highest contribution per unit (€31.80) but the lowest contribution per gram of the scarce resource (€6.81), because it consumes far more grams per unit than the other products. The correct basis for prioritising production under a resource constraint is always contribution per unit of the limiting factor, not contribution per unit of product. HydraGlow and ClearSkin are close (€9.56 vs €9.36 per gram) but HydraGlow still ranks marginally ahead.

Full Model Answer (continued) – Step 3: allocate the 70,000 grams and finalise the production plan**Working – allocation of available extract (70,000 grams) in rank order**

Rank	Product	Grams needed for full demand	Grams allocated	Cumulative grams used	Units produced
1st	Radiance	30,000	30,000	30,000	18,000 (full demand)
2nd	HydraGlow	32,000	32,000	62,000	12,000 (full demand)
3rd	ClearSkin	25,000	8,000 (remaining balance)	70,000	2,400 (partial)
4th	YouthRenew	18,667	0	70,000	0

ClearSkin partial production: After meeting full demand for Radiance (30,000 grams) and HydraGlow (32,000 grams), only 70,000 – 62,000 = 8,000 grams remain. At 3.333 grams per unit, this supports $8,000 \div 3.333 = 2,400$ units of ClearSkin, out of the 7,500 units demanded — leaving a shortfall of 5,100 ClearSkin units, and no capacity remaining at all for YouthRenew.

Resulting profit-maximising production plan and contribution

Product	Units to produce	Contribution/unit (€)	Total contribution (€)
Radiance	18,000	26.30	473,400
HydraGlow	12,000	25.50	306,000
ClearSkin	2,400	31.20	74,880
YouthRenew	0	31.80	0
Total contribution			854,280

Assumptions made

1. Fixed costs (the fixed portions of production overheads, marketing, and other costs) are assumed to remain unchanged regardless of the production mix, so maximising total contribution is equivalent to maximising total profit. **2.** The essential plant extract is assumed to be the only binding constraint on production — other resources (labour, machine capacity) are assumed sufficient to produce whatever mix the extract allows. **3.** Products are assumed divisible (partial units of ClearSkin can be produced/sold) for the purposes of fully utilising the scarce resource; in practice, ClearSkin production would likely be rounded to a whole number of saleable units (2,400 exactly, as it happens, requires no rounding here).

Brief commentary on the plan

The optimal plan fully satisfies demand for the two most extract-efficient products (Radiance and HydraGlow) but leaves a significant 5,100-unit shortfall in ClearSkin and eliminates YouthRenew production entirely. Given that YouthRenew is PureGlow's premium, highest-priced product (€150) and ClearSkin also carries a strong margin, Ms. White should recognise the plan is deliberately allocating the extract shortage away from the most resource-intensive products, which may need to be managed carefully from a customer relationship and premium-brand-positioning perspective, even though it is the profit-maximising response to the constraint as it stands.

Mark Allocation

Marks	Criteria
17-20	Limiting factor correctly confirmed; variable cost, contribution per unit, and contribution per gram all correctly calculated; correct ranking; correct allocation of the 70,000 grams giving the profit-maximising plan; clear assumptions and relevant commentary.
12-16	Substantially correct approach with a minor computational error, e.g. in the variable cost split or ranking of the two close products.
6-11	Limiting factor approach understood in principle but ranked by contribution per unit rather than per unit of limiting factor, or allocation contains significant errors.
0-5	Limited understanding of limiting factor analysis; minimal correct workings.

Question 2(b)**12 marks**

Outline and briefly explain FOUR potential actions that PureGlow Skincare Ltd could take to ensure they meet the total demand for their skincare products in 2025.

Full Model Answer**1. Source alternative or additional suppliers**

PureGlow could seek out additional suppliers of essential plant extracts, potentially in different geographic regions less affected by the anticipated shortage, to increase total available supply above 70,000 grams.

2. Reformulate products to reduce extract usage

Investing in product reformulation or R&D to reduce the quantity of essential plant extract required per unit (without significantly compromising product quality or efficacy) would increase the number of units that can be produced from the same total supply, easing the constraint across all four products.

3. Secure long-term or advance supply agreements

Entering into long-term contracts or advance purchase agreements with existing suppliers could help secure priority access to available extract supply ahead of competitors also facing the same global shortage, reducing the risk of supply disruption.

4. Identify substitute materials

PureGlow could research and qualify substitute or synthetic ingredients that could partially replace the scarce essential plant extracts in some formulations, freeing up the remaining natural extract supply for products where substitution is not feasible or would compromise the 'organic' positioning.

Other acceptable actions: Candidates may alternatively discuss: raising prices on the most extract-intensive products to manage/ration demand down towards available supply; building buffer stock of extract in advance where feasible; or outsourcing production of some products to a third party with independent access to extract supply. Any four well-explained, distinct actions receive full credit.

Mark Allocation

Marks	Criteria
10-12	Four distinct, well-explained actions given, each a genuine and practical response to a raw-material supply constraint.
7-9	Four actions identified with reasonable explanation, or three actions explained in excellent depth.
4-6	Two to three actions identified with limited explanation.
0-3	One or no relevant actions identified.

Question 2(c)**9 marks**

Outline and briefly explain the THREE criteria that must be met for any cost (or benefit) to be considered as relevant for decision-making purposes.

Full Model Answer**1. It must be a future cost or benefit**

A relevant cost or benefit must relate to something that will occur in the future, as a consequence of the decision being made now. Costs already incurred (sunk costs, see part (d)) cannot be changed by the decision and are therefore never relevant.

2. It must differ between the alternatives being considered (it must be incremental/differential)

A cost or benefit is only relevant if it changes depending on which course of action is chosen. Any cost or benefit that is identical regardless of which alternative is selected has no bearing on the decision and can be ignored, even if it is a genuine future cash flow.

3. It must represent an actual cash flow

Relevant costs and benefits must correspond to real cash inflows or outflows. Purely accounting entries or allocations — such as depreciation charges, or absorbed fixed overhead that would be incurred regardless of the decision — do not represent genuine cash flows arising from the decision and are not relevant, even though they may appear in the accounting records.

Mark Allocation

Marks	Criteria
8-9	All three criteria correctly identified and clearly explained, using accurate terminology (future, incremental/differential, cash flow).
5-7	All three criteria identified with reasonable explanation, or two explained in excellent depth.
2-4	One or two criteria identified with limited explanation.
0-1	No relevant criteria identified.

Question 2(d)**9 marks**

Outline and briefly explain what is meant by each of the following terms: (i) Joint products; (ii) Sunk cost; (iii) Opportunity cost.

Full Model Answer**(i) Joint products**

Joint products are two or more distinct products that are produced simultaneously from the same input, process, or raw material, up to a 'split-off point' beyond which they become separately identifiable and may require further individual processing. An example would be a single batch of essential plant extract processing yielding several different active compounds that PureGlow could use across its Radiance, HydraGlow, ClearSkin and YouthRenew product lines.

(ii) Sunk cost

A sunk cost is a cost that has already been incurred in the past and cannot be recovered, avoided, or changed by any current or future decision. Because it does not differ between the alternatives being considered, a sunk cost is irrelevant to decision-making and should be ignored (see part (c)).

(iii) Opportunity cost

An opportunity cost is the value of the benefit forgone by choosing one course of action over the next best available alternative. It does not appear anywhere in the accounting records, but represents a genuine economic sacrifice and is therefore highly relevant to decision-making — for example, the contribution PureGlow forgoes by using its limited extract supply on ClearSkin rather than YouthRenew is an opportunity cost of the production plan chosen in part (a).

Mark Allocation

Marks	Criteria
8-9	All three terms accurately and fully defined, with a relevant example given for at least one, ideally tied to the PureGlow scenario.
5-7	All three terms defined with reasonable accuracy; examples missing or generic.
2-4	One or two terms defined accurately.
0-1	No meaningful definitions provided.

QUESTION 2 TOTAL — 50 MARKS

QUESTION 3**EcoGlow Ltd**

Total: 50 marks

EcoGlow Ltd manufactures the "Breathe+" air purifier. The newly appointed Managing Director wants to assess 2024 trading performance, comparing budget to actual, where actual production and sales were 8% lower than the original 18,500-unit plan due to supply chain disruption. Candidates are required to (a) construct a flexible budget and calculate flexible budget variances and volume variances, (b) evaluate EcoGlow's financial performance based on that analysis, and (c) outline four criticisms of budgeting.

Part	Requirement	Marks
(a)	Flexible budget, flexible budget variances (FBV) and volume variances (VV), each marked favourable or unfavourable	24
(b)	Evaluate 2024 financial performance and advise Mr. O'Connor	14
(c)	Four criticisms of budgeting	12

Question 3(a)

24 marks

EcoGlow Ltd, 2024, Breathe+ air purifier: Budget (based on 18,500 units) and Actual figures given for sales revenue, variable costs (direct materials, direct labour, manufacturing overheads, marketing, distribution) and fixed costs (insurance, salaries, rent, utilities, other). Actual production and sales were 8% lower than the 18,500-unit plan. (a) Construct a flexible budget for 2024 in relation to Breathe+. Then, using the flexible budget, master budget and actual amounts, calculate flexible budget variances (FBVs) and volume variances (VVs) for each budgetary item, indicating whether each variance is favourable (F) or unfavourable (U).

Full Model Answer

Working - actual volume and per-unit budget rates

Item	Value
Master budget volume	18,500 units
Actual volume (8% lower than plan)	17,020 units
Budgeted selling price per unit (€1,850,000 ÷ 18,500)	€100.00

Method: The flexible budget re-states the master budget's variable items (sales revenue and each variable cost) at the *actual* volume of 17,020 units, using the *budgeted* rate per unit for each. Fixed costs are unchanged in the flexible budget, since by definition they do not vary with volume within the relevant range. The flexible budget variance (FBV) then isolates genuine spending/price/efficiency differences (actual vs flexible budget, both at the same, actual volume), while the volume variance (VV) isolates the effect of the volume shortfall alone (flexible budget vs master budget, both at budgeted rates).

Full Model Answer (continued) – Flexible budget and variances: variable items**Flexible budget, FBV and VV – sales revenue and variable costs**

Item	Master budget (€)	Flexible budget (€)	Actual (€)	FBV (Actual – Flex)	VV (Flex – Master)
Sales revenue	1,850,000	1,702,000	1,665,000	(37,000) U	(148,000) U
Direct materials	580,000	533,600	540,000	(6,400) U	46,400 F
Direct labour	440,000	404,800	390,000	14,800 F	35,200 F
Manufacturing overheads	190,000	174,800	175,000	(200) U	15,200 F
Marketing	90,000	82,800	82,500	300 F	7,200 F
Distribution	70,000	64,400	63,500	900 F	5,600 F

Sign convention for costs vs revenue: For sales revenue, a positive figure (actual/flexible exceeding the comparison) is favourable; for costs, a **lower** actual/flexible figure (i.e. a negative variance in the Actual – Flex or Flex – Master calculation) is favourable, since it means less was spent than expected. All cost variances above are labelled F or U according to this convention, not simply by the arithmetic sign of the subtraction.

Contribution

Item	Master budget (€)	Flexible budget (€)	Actual (€)	FBV	VV
Contribution	480,000	441,600	413,500	(28,100) U	(38,400) U

Note on the actual figures as given: The individual actual variable cost line items given in the exam paper sum to €1,251,000, which is €500 less than the €1,251,500 'Total variable costs' figure stated in the same table; the stated actual contribution of €413,500 and actual net profit of €160,000 are consistent with the €1,251,500 total, not with the sum of the individual lines, indicating a minor rounding/typing inconsistency in one of the individual actual cost figures as issued. This model answer uses the paper's own stated Contribution and Net Profit subtotals as authoritative, since the final net profit figure only reconciles correctly on that basis; candidates working consistently from the individual line items (giving a contribution of €414,000 and net profit of €160,500) receive full own-figure credit.

Full Model Answer (continued) – Fixed costs and net profit**Flexible budget, FBV and VV – fixed costs**

Item	Master budget (€)	Flexible budget (€)	Actual (€)	FBV (Actual – Flex)	VV (Flex – Master)
Insurance	47,000	47,000	56,000	(9,000) U	0
Salaries	125,000	125,000	140,000	(15,000) U	0
Rent	28,000	28,000	26,000	2,000 F	0
Utilities	20,000	20,000	22,500	(2,500) U	0
Other fixed costs	8,000	8,000	9,000	(1,000) U	0
Total fixed costs	228,000	228,000	253,500	(25,500) U	0

Why fixed costs have no volume variance: Fixed costs are, by definition, unaffected by the volume of activity within the relevant range, so the flexible budget figure for every fixed cost line is identical to the master budget figure — the volume variance for every fixed cost item is therefore zero. Any difference between budget and actual for fixed costs is entirely a flexible budget (spending) variance.

Net profit

Item	Master budget (€)	Flexible budget (€)	Actual (€)	FBV	VV
Net profit	252,000	213,600	160,000	(53,600) U	(38,400) U

Overall reconciliation: The total variance in net profit (Actual €160,000 – Master budget €252,000 = €(92,000) unfavourable) is fully explained by the sum of the flexible budget variance (€(53,600) unfavourable) and the volume variance (€(38,400) unfavourable): €(53,600) + €(38,400) = €(92,000).

Mark Allocation

Marks	Criteria
20–24	Actual volume correctly derived (17,020 units); flexible budget correctly constructed for all variable items at budgeted per-unit rates; fixed costs correctly held constant; all FBVs and VVs correctly calculated and correctly labelled F/U; net profit reconciliation correct.
14–19	Substantially correct with minor errors in one or two line items, or F/U labelling inconsistently applied.
7–13	Flexible budget concept understood but significant errors in the FBV/VV split, or fixed costs incorrectly flexed with volume.
0–6	Limited understanding of flexible budgeting; minimal correct workings.

Question 3(b)**14 marks**

Based upon your answer to (a), briefly evaluate EcoGlow Ltd.'s financial performance during 2024 in relation to "Breathe+", offering whatever advice to Mr. O'Connor that you consider appropriate.

Full Model Answer**Headline result**

Actual net profit of €160,000 fell €92,000 (36.5%) short of the original €252,000 budget. Simply comparing actual to the master budget, however, would unfairly blend two very different issues together — the variance analysis in part (a) allows these to be separated and assessed on their own merits.

The volume variance (€(38,400) unfavourable) is largely outside management's immediate control

This portion of the shortfall stems directly from the 8% supply-chain-driven fall in production and sales volume, which the question describes as an unexpected disruption. While EcoGlow's supply chain resilience could reasonably be questioned going forward, this element of the shortfall does not, on its own, reflect poor operational management of the business during the period.

The flexible budget variance (€(53,600) unfavourable) is more concerning

This is the larger of the two components and represents genuine spending and efficiency issues *after* already adjusting for the lower volume achieved — i.e. even allowing for selling fewer units, the business still spent more than it should have. Within this: direct materials, manufacturing overheads and fixed costs overall were unfavourable, but direct labour, marketing and distribution were all favourable (better cost control or lower prices than budgeted) at the actual volume. The most significant individual overruns are in **salaries** (€(15,000) unfavourable) and **insurance** (€(9,000) unfavourable), which together account for the majority of the fixed cost overspend and merit specific investigation — for example, whether unbudgeted pay increases, new hires, or a premium increase drove these results.

Advice to Mr. O'Connor

Investigate the salary and insurance overruns specifically, as these are the largest controllable drivers of the shortfall. Recognise that direct material, labour, and variable overhead cost control was actually reasonably good once volume is properly accounted for — the story is not one of uniformly poor performance. Separately, given that volume disruption contributed a similarly large share of the total shortfall, EcoGlow should review the resilience of its supply chain to reduce the risk of a repeat volume shortfall in future periods.

Mark Allocation

Marks	Criteria
12-14	Correctly separates the volume-driven shortfall (largely uncontrollable) from the flexible budget variance (largely controllable); identifies salaries and insurance as the key drivers of the unfavourable FBV; gives specific, evidence-based advice referencing the figures from part (a).
8-11	Good evaluation distinguishing FBV from VV with reasonable identification of key drivers; advice more generic.
4-7	Some relevant commentary but does not clearly distinguish the two variance types, or lacks specific reference to individual line items.
0-3	Minimal or no meaningful evaluation.

Question 3(c)**12 marks**

Outline and briefly explain FOUR criticisms associated with the use of budgeting that Mr. O'Connor should be aware of.

Full Model Answer**1. Time-consuming and costly to prepare**

Preparing a detailed annual budget can consume significant management time and organisational resources, time that could otherwise be spent on value-adding activities. For a business the size of EcoGlow, this administrative burden must be weighed against the benefits budgeting provides.

2. Rigid/inflexible once set, especially without a flexible budgeting adjustment

A fixed master budget quickly becomes an inappropriate basis for comparison once actual conditions change, as illustrated directly by this question: comparing EcoGlow's actual results straight against the original 18,500-unit master budget would produce a misleading picture, requiring the flexible budget adjustment in part (a) to give a fair, like-for-like comparison.

3. Can encourage dysfunctional behaviour and budgetary slack

Where managers are evaluated against meeting budget targets, this can create an incentive to deliberately understate achievable revenue or overstate necessary costs when the budget is being set (building in 'slack' to make future targets easier to hit), or to spend up to the full remaining budget near the end of a period purely to avoid a reduced allocation next year, even where the spending is not genuinely needed.

4. Backward-looking and can perpetuate inefficiency

Particularly where an incremental budgeting approach is used (see Question 1(c)), budgets are often built from the prior period's figures with minor adjustments, rather than being rigorously reassessed from first principles. This risks simply carrying forward existing inefficiencies year after year, and can encourage a short-term focus on meeting the current year's budget target at the expense of longer-term strategic objectives.

Other acceptable criticisms: Candidates may alternatively discuss: budgets can become outdated quickly in a fast-changing or uncertain trading environment; they may not clearly link departmental targets to overall organisational strategy or value creation; or they may create silo behaviour between departments competing for a limited budget allocation. Any four well-explained, distinct criticisms receive full credit.

Mark Allocation

Marks	Criteria
10-12	Four distinct, well-explained criticisms given, each a genuine and recognised limitation of budgeting; ideally at least one linked to the specific issues encountered in this question.
7-9	Four criticisms identified with reasonable explanation, or three explained in excellent depth.
4-6	Two to three criticisms identified with limited explanation.
0-3	One or no relevant criticisms identified.

QUESTION 3 TOTAL — 50 MARKS